

Q1.

A curve passes through the point (1, 3) and satisfies the differential equation

$$\frac{dy}{dx} = \frac{x}{1+x^3}$$

Starting at the point (1, 3), use a step-by-step method with a step length of 0.1 to estimate the value of y at $x = 1.2$. Give your answer to four decimal places.

(Total 5 marks)

Q2.

It is given that $y(x)$ satisfies the differential equation

$$\frac{dy}{dx} = f(x, y)$$

where

$$f(x, y) = \sqrt{2x+y}$$

and

$$y(3) = 5$$

(a) Use the Euler formula

$$y_{r+1} = y_r + hf(x_r, y_r)$$

with $h = 0.2$, to obtain an approximation to $y(3.2)$, giving your answer to four decimal places.

(3)

(b) Use the formula

$$y_{r+1} = y_{r-1} + 2hf(x_r, y_r)$$

with your answer to part (a), to obtain an approximation to $y(3.4)$, giving your answer to three decimal places.

(3)

(Total 6 marks)

Q3.

The function $y(x)$ satisfies the differential equation

$$\frac{dy}{dx} = f(x, y)$$

$$\text{where } f(x, y) = \frac{y-x}{y^2+x}$$

and

$$y(1) = 2$$

(a) Use the Euler formula

$$y_{r+1} = y_r + hf(x_r, y_r)$$

with $h = 0.1$, to obtain an approximation to $y(1.1)$.

(3)

(b) Use the formula

$$y_{r+1} = y_{r-1} + 2hf(x_r, y_r)$$

with your answer to part (a), to obtain an approximation to $y(1.2)$, giving your answer to three decimal places.

(3)

(Total 6 marks)

Q4.

A curve passes through the point (2, 3) and satisfies the differential equation

$$\frac{dy}{dx} = \frac{1}{\sqrt{2+x}}$$

Starting at the point (2, 3), use a step-by-step method with a step length of 0.5 to estimate the value of y at $x = 3$. Give your answer to four decimal places.

(Total 5 marks)

Q5.

The function $y(x)$ satisfies the differential equation

$$\frac{dy}{dx} = f(x, y)$$

where

$$f(x, y) = x \ln(2x + y)$$

and

$$y(3) = 2$$

(a) Use the Euler formula

$$y_{r+1} = y_r + hf(x_r, y_r)$$

with $h = 0.1$, to obtain an approximation to $y(3.1)$, giving your answer to four decimal places.

(3)

(b) Use the improved Euler formula

$$y_{r+1} = y_r + \frac{1}{2}(k_1 + k_2)$$

where $k_1 = hf(x_r, y_r)$ and $k_2 = hf(x_r + h, y_r + k_1)$ and $h = 0.1$, to obtain an approximation to $y(3.1)$, giving your answer to four decimal places.

(5)

Q6.

A curve passes through the point (1, 3) and satisfies the differential equation

$$\frac{dy}{dx} = 1 + x^3$$

Starting at the point (1, 3), use a step-by-step method with a step length of 0.1 to estimate the y-coordinate of the point on the curve for which $x = 1.3$. Give your answer to three decimal places.

(No credit will be given for methods involving integration.)

(Total 6 marks)

Q7.

The function $y(x)$ satisfies the differential equation

$$\frac{dy}{dx} = f(x, y)$$

where $f(x, y) = x + 3 + \sin y$

and $y(1) = 1$

(a) Use the Euler formula

$$y_{r+1} = y_r + hf(x_r, y_r)$$

with $h = 0.1$, to obtain an approximation to $y(1.1)$, giving your answer to four decimal places.

(3)

(b) Use the formula

$$y_{r+1} = y_{r-1} + 2hf(x_r, y_r)$$

with your answer to part (a), to obtain an approximation to $y(1.2)$, giving your answer to three decimal places.

(3)

(Total 6 marks)

Q8.

A curve passes through the point (0, 1) and satisfies the differential equation

$$\frac{dx}{dy} = \sqrt{1 + x^2}$$

Starting at the point (0, 1), use a step-by-step method with a step length of 0.2 to estimate the value of y at $x = 0.4$. Give your answer to five decimal places.

Q9.

The function $y(x)$ satisfies the differential equation

$$\frac{dy}{dx} = f(x, y)$$

$$f(x, y) = \frac{x^2 + y^2}{x + y}$$

where

and $y(1) = 3$

(a) Use the Euler formula

$$y_{r+1} = y_r + hf(x_r, y_r)$$

with $h = 0.2$, to obtain an approximation to $y(1.2)$.

(3)

(b) Use the improved Euler formula

$$y_{r+1} = y_r + \frac{1}{2}(k_1 + k_2)$$

where $k_1 = hf(x_r, y_r)$ and $k_2 = hf(x_r + h, y_r + k_1)$ and $h = 0.2$, to obtain an approximation to $y(1.2)$, giving your answer to four decimal places.

(5)

(Total 8 marks)

Q10.

The function $y(x)$ satisfies the differential equation

$$\frac{dy}{dx} = f(x, y)$$

where

$$f(x, y) = \sqrt{x^2 + y + 1}$$

and

$$y(3) = 2$$

(a) Use the Euler formula

$$y_{r+1} = y_r + hf(x_r, y_r)$$

with $h = 0.1$, to obtain an approximation to $y(3.1)$, giving your answer to four decimal places.

(3)

(b) Use the formula

$$y_{r+1} = y_{r-1} + 2hf(x_r, y_r)$$

with your answer to part (a), to obtain an approximation to $y(3.2)$, giving your answer to three decimal places.

(3)

(Total 6 marks)

Q11.

A curve satisfies the differential equation

$$\frac{dy}{dx} = 2^x$$

Starting at the point (1, 4) on the curve, use a step-by-step method with a step length of 0.01 to estimate the value of y at $x = 1.02$. Give your answer to six significant figures.

(Total 5 marks)

Q12.

The function $y(x)$ satisfies the differential equation

$$\frac{dy}{dx} = f(x, y)$$

where

$$f(x, y) = x^2 - y^2$$

and

$$y(2) = 1$$

- (a) Use the Euler formula

$$y_{r+1} = y_r + hf(x_r, y_r)$$

with $h = 0.1$, to obtain an approximation to $y(2.1)$.

(3)

- (b) Use the formula

$$y_{r+1} = y_{r-1} + 2hf(x_r, y_r)$$

with your answer to part (a), to obtain an approximation to $y(2.2)$.

(3)

(Total 6 marks)

Q13.

The function $y(x)$ satisfies the differential equation

$$\frac{dy}{dx} = f(x, y)$$

where $f(x, y) = \ln(1 + x^2 + y)$

and $y(1) = 0.6$

(a) Use the Euler formula

$$y_{r+1} = y_r + h f(x_r, y_r)$$

with $h = 0.05$, to obtain an approximation to $y(1.05)$, giving your answer to four decimal places.

(3)

(b) Use the improved Euler formula

$$y_{r+1} = y_r + \frac{1}{2}(k_1 + k_2)$$

where $k_1 = h f(x_r, y_r)$ and $k_2 = h f(x_r + h, y_r + k_1)$ and $h = 0.05$, to obtain an approximation to $y(1.05)$, giving your answer to four decimal places.

(6)

(Total 9 marks)

Q14.

The function $y(x)$ satisfies the differential equation

$$\frac{dy}{dx} = f(x, y)$$

where

$$f(x, y) = \sqrt{x^2 + y^2 + 3}$$

and

$$y(1) = 2$$

(a) Use the Euler formula

$$y_{r+1} = y_r + hf(x_r, y_r)$$

with $h = 0.1$, to obtain an approximation to $y(1.1)$, giving your answer to four decimal places.

(3)

(b) Use the improved Euler formula

$$y_{r+1} = y_r + \frac{1}{2}(k_1 + k_2)$$

where $k_1 = hf(x_r, y_r)$ and $k_2 = hf(x_r + h, y_r + k_1)$ and $h = 0.1$, to obtain an approximation to $y(1.1)$, giving your answer to four decimal places.

(6)

(Total 9 marks)

Q15.

- (a) The function $y(x)$ satisfies the differential equation

$$\frac{dy}{dx} = f(x, y)$$

where

$$f(x, y) = x \ln x + \frac{y}{x}$$

and

$$y(1) = 1$$

- (i) Use the Euler formula

$$y_{r+1} = y_r + h f(x_r, y_r)$$

with $h = 0.1$, to obtain an approximation to $y(1.1)$.

(3)

- (ii) Use the formula

$$y_{r+1} = y_{r-1} + 2h f(x_r, y_r)$$

with your answer to part (a)(i) to obtain an approximation to $y(1.2)$, giving your answer to three decimal places.

(4)

- (b) (i) Show that $\frac{1}{x}$ is an integrating factor for the first-order differential equation

$$\frac{dy}{dx} - \frac{1}{x}y = x \ln x$$

(3)

- (ii) Solve this differential equation, given that $y = 1$ when $x = 1$.

(6)

- (iii) Calculate the value of y when $x = 1.2$, giving your answer to three decimal places.

(1)

(Total 17 marks)

Q16.

A curve satisfies the differential equation

$$\frac{dy}{dx} = \log_{10} x$$

Starting at the point (2, 3) on the curve, use a step-by-step method with a step length

of 0.2 to estimate the value of y at $x = 2.4$. Give your answer to three decimal places.

(Total 6 marks)

Q17.

The function $y(x)$ satisfies the differential equation

$$\frac{dy}{dx} = f(x, y)$$

where $f(x, y) = \frac{x^2 + y^2}{xy}$
and $y(1) = 2$

(a) Use the Euler formula

$$y_{r+1} = y_r + hf(x_r, y_r)$$

with $h = 0.1$, to obtain an approximation to $y(1.1)$.

(3)

(b) Use the improved Euler formula

$$y_{r+1} = y_r + \frac{1}{2}(k_1, k_2)$$

where $k_1 = hf(x_r, y_r)$ and $k_2 = hf(x_r + h, y_r + k_1)$ and $h = 0.1$, to obtain an approximation to $y(1.1)$, giving your answer to four decimal places.

(6)

(Total 9 marks)

Q18.

A curve satisfies the differential equation

$$\frac{dy}{dx} = \sin 2x$$

where the angle $2x$ is measured in **radians**.

Starting at the point $(0.5, 1)$ on the curve, use a step-by-step method with a step length of 0.1 to estimate the value of y at $x = 0.7$. Give your answer to three significant figures.

(Total 6 marks)